

1. Frontend (The Client Layer):

Technologies:

- **HTML5:** Used for the semantic structure of the Hospital Booking Form and the Appointment Table.
- **CSS3:** Used for the responsive design, modern UI/UX, and professional color palette.
- **JavaScript (ES6+):** Handles the "Fetch API" calls to send data to the server and dynamically update the table without reloading.

AI Tools Used:

- **ChatGPT:** Generated the initial HTML structure.
 - **Claude:** Optimized the CSS for a clean, medical-themed design.
 - **Lovable dev:** Streamlined the frontend JavaScript logic.
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2. Backend (The Logic Layer):

Technologies:

- **Node.js:** The core runtime environment that allows the server to handle multiple requests.
- **Express.js:** The framework used to build the REST API. It provides the endpoints (POST /api/appointments and GET /api/appointments).
- **Mongoose:** Used as the Schema-based solution to model the application data and connect Node.js to MongoDB.
- **CORS:** Middleware used to allow the frontend (running on your browser) to talk to the backend safely.

AI Tools Used:

- **Cursor AI:** Used to build the entire folder structure, server.js, and the backend routes based on the prompts provided.

3. Database (The Storage Layer):

Technologies:

- **MongoDB:** A NoSQL database that stores appointment records as flexible, JSON-like documents.
- **Atlas/Local Server:** The hosting environment where the hospital_db is stored.

- Management Tools:

MongoDB Compass:

The official GUI used to browse, query, and verify the data inside the appointments collection.

4. Development Tools (The Environment):

- IDE: Cursor Code Editor (An AI-integrated editor used for writing, debugging, and merging code).
- Package Manager: npm (Node Package Manager) used to install and manage the project dependencies.
- Server Tool: Live Server (used for viewing the HTML frontend in real-time).
- Version Control/Review: Cursor Diff View (used to review and accept AI-suggested code changes).